

- 6 Two 60 W microwave point sources, which are placed 7.0 m apart as shown in Fig. 6.1, emit microwaves in all directions. The microwave from the two sources are always in phase. Position M is the midpoint between the two microwave sources.

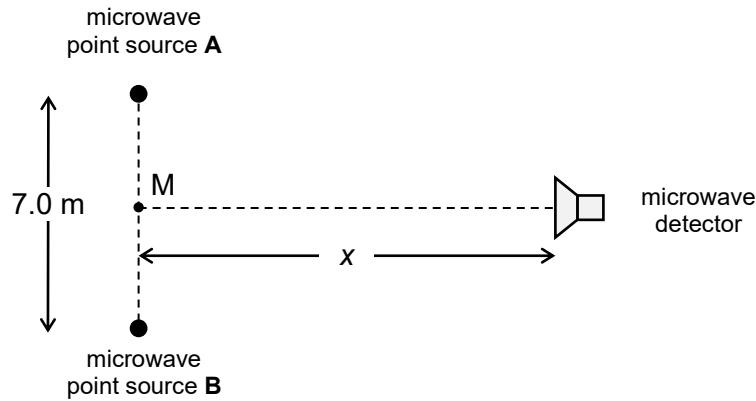


Fig. 6.1

- (a) A microwave detector is placed a distance x from position M. It has an aperture of 5.0 cm. The frequency of the microwave is set to 75 GHz.

Determine the maximum distance $x_{\max}$ such that the detector is just able to resolve the two point sources.

$$x_{\max} = \dots\dots\dots \text{ m [3]}$$

- (b) The frequency of the microwave sources is adjusted to 0.75 GHz. The microwave detector is placed at position P as shown in Fig. 6.2.

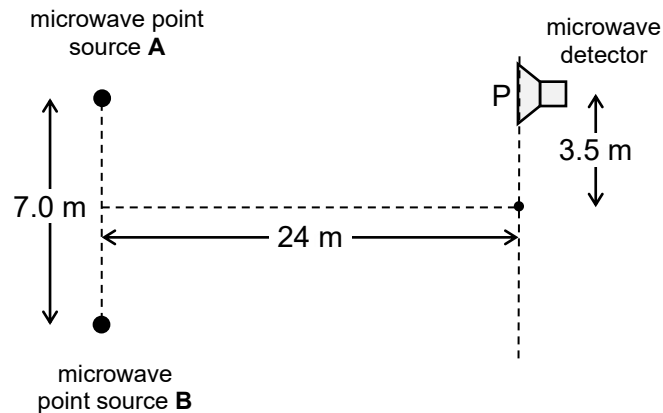


Fig. 6.2

- (i) Determine the intensity that is measured by the detector,
1. if only emitter A is switched on.

intensity due to A = W m^{-2} [1]

2. if only emitter B is switched on.

intensity due to B = W m^{-2} [2]

- (ii) Hence, determine the ratio of $\frac{\text{amplitude due to source B}}{\text{amplitude due to source A}}$ at P.

ratio = [2]

- (iii) Determine the phase difference between the microwaves from source A and source B at P.

phase difference = rad [2]

- (iv) Hence, determine the resultant intensity of the microwave at P.

resultant intensity = W m^{-2} [2]

- (c) The two sources need to produce coherent microwaves for the calculations in part (b) to be valid.

State and explain what would be detected at P if the microwaves from the two sources are not coherent.

.....

.....

.....

.....

..... [2]